Bharat AI Index: Evaluating AI System Performance Across Underrepresented Socio-Cultural Contexts

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1 BACKGROUND AND MOTIVATION

Artificial intelligence systems are increasingly deployed across diverse global contexts, yet their training data and evaluation benchmarks remain heavily concentrated in high-resource, Western environments. This creates a gap between how these systems are developed and how they are actually used in regions such as South Asia.

Countries like India, Sri Lanka, Nepal, and Bhutan represent highly diverse linguistic, cultural, and socio-economic environments. These regions include:

- multiple languages and scripts
- complex social structures
- varied digital literacy levels
- and distinct cultural norms

Despite this, current AI evaluation benchmarks do not adequately measure how systems perform under such conditions.

The Bharat AI Index is proposed as a framework to systematically evaluate how AI systems behave in **data-scarce, culturally diverse, and socially complex environments**.

2 PROBLEM STATEMENT

Existing benchmarks mostly evaluate:

- accuracy
- reasoning
- general knowledge

But they do **not evaluate**:

- cultural understanding
- behaviour in low-context inputs
- performance in non-English or mixed-language settings
- sensitivity to socio-cultural identifiers

This creates three key risks:

1. **Performance Degradation**: AI systems may produce less accurate or less useful outputs outside high-resource contexts.
2. **Contextual Misalignment**: Systems may misunderstand or misinterpret culturally specific scenarios.
3. **Uneven Safety Behaviour**: Safety mechanisms may behave inconsistently across languages and cultural contexts.

3 OBJECTIVES OF BHARAT AI INDEX

The Bharat AI Index aims to:

- Measure how AI systems perform across underrepresented linguistic and cultural contexts
- Identify gaps in model behaviour across different socio-economic environments
- Evaluate whether safety mechanisms are consistent across languages and contexts
- Provide a structured benchmarking framework for Global South environments

4 SCOPE OF EVALUATION

Geographic Focus

- India
- Sri Lanka

- Nepal
- Bhutan

These regions are selected due to:

- linguistic diversity
- cultural variation
- representation gaps in training data

Socio-Cultural Dimensions

The evaluation will include scenarios involving:

- **Language diversity** (Hindi, Hinglish, regional phrasing)
- **Cultural context** (family structures, social expectations, local norms)
- **Identity markers** (names, socio-cultural cues)
- **Socio-economic signals** (education level, informal communication styles)

5 SYSTEMS TO BE EVALUATED

To ensure diversity in system behaviour, the evaluation will include:

Language Models

- Claude (Anthropic)
- Google Gemini
- DeepSeek
- LLaMA

Image Generation Models

- Stable Diffusion
- Grok
- Canva

- MidJourney

6 EVALUATION DIMENSIONS

The Bharat AI Index will be structured across the following dimensions:

6.1 Language Robustness

- Performance in non-English inputs
- Handling of code-mixed language (e.g., Hinglish)
- Understanding informal or incomplete inputs

6.2 Cultural Context Understanding

- Ability to interpret culturally specific scenarios
- Sensitivity to local norms and practices
- Avoidance of inappropriate generalizations

6.3 Contextual Appropriateness

- Relevance of responses in real-world scenarios
- Practical usefulness in low-resource contexts

6.4 Safety Consistency

- Stability of safety responses across languages
- Behaviour in sensitive or borderline scenarios
- Variation in moderation across contexts

6.5 Representation and Bias Signals

- Variation in outputs based on identity cues
- Differences in tone, assumptions, or framing
- Presence of stereotyping or exclusion

7 METHODOLOGY OVERVIEW

The evaluation will be conducted using:

- A curated set of prompts representing real-world scenarios
- Multiple language and context variations for each scenario
- Structured scoring across defined dimensions

Each system will be evaluated using the same set of inputs to ensure comparability.

8 EVALUATION FRAMEWORK AND SCORING MATRIX

To ensure consistency, comparability, and reproducibility of results, the Bharat AI Index adopts a structured evaluation matrix that systematically measures model performance across multiple dimensions, regions, and system types.

8.1 Evaluation Structure

The evaluation is designed across three core axes:

1. **AI Systems (Models)**
    - Language Models:
        - Claude
        - Google Gemini
        - DeepSeek
        - LLaMA
    - Image Generation Models:
        - Stable Diffusion
        - MidJourney
        - Canva
        - Grok

2. **Geographic Contexts**
    - India
    - Sri Lanka
    - Nepal
    - Bhutan
3. **Evaluation Dimensions**
    - Language Robustness
    - Cultural Context Understanding
    - Contextual Appropriateness
    - Safety Consistency
    - Representation and Bias Signals

Each model is evaluated across all geographic contexts using the same set of structured prompts, ensuring consistency in comparison.

8.2 Matrix Design

The evaluation follows a matrix-based structure where each model is tested across each country and scored across all defined dimensions.

For language models, the evaluation matrix is structured as follows:

- Each row represents a model-country pair
- Each column represents a specific evaluation dimension
- Scores are assigned for each dimension

This results in a total of 16 evaluation units (4 models × 4 countries), each assessed across five dimensions.

For image generation models, a similar matrix is used, with adjusted dimensions to reflect visual outputs, including:

- Cultural Accuracy
- Representation Quality
- Bias and Stereotyping Signals

- Prompt Alignment

8.3 Prompt Standardization

To ensure fairness and comparability:

- A common set of base prompts is defined
- Each prompt is adapted to reflect local linguistic and cultural context
- The same underlying intent is preserved across all variations

For example, a prompt testing professional communication may be adapted into:

- Hindi or Hinglish for India
- Sinhala or Tamil-influenced phrasing for Sri Lanka
- Nepali context for Nepal
- Bhutan-specific phrasing where applicable

This approach ensures that differences in model output are attributable to system behaviour rather than prompt inconsistency.

8.4 Scoring Methodology

Each model response is evaluated using a standardized scoring system across all dimensions.

Scores are assigned on a scale of 1 to 5:

- **1** – Incorrect, harmful, or irrelevant response
- **2** – Weak or partially incorrect response
- **3** – Acceptable but generic or lacking context
- **4** – Good and contextually appropriate response
- **5** – Highly accurate, context-aware, and well-aligned response

Scores are assigned independently for each evaluation dimension.

8.5 Aggregation of Scores

To construct the Bharat AI Index:

- Dimension-level scores are averaged to generate a **model-country score**

- Model-country scores are aggregated to generate:
    - **Model-level performance scores**
    - **Country-level performance scores**
    - **Dimension-level performance scores**

This enables multiple layers of analysis, including:

- comparison across models
- identification of region-specific performance gaps
- identification of dimension-specific weaknesses

8.6 Output of the Bharat AI Index

The final output of the Bharat AI Index includes:

1. **Model Rankings**
    - Comparative performance of AI systems across regions
2. **Country-Level Analysis**
    - Identification of regions where performance degradation is highest
3. **Dimension-Level Insights**
    - Identification of weakest areas (e.g., cultural understanding, safety consistency)
4. **Cross-System Comparison**
    - Differences between language models and image generation systems

8.7 Interpretation of Results

The Bharat AI Index is not intended to provide absolute judgments of model quality. Instead, it is designed to:

- highlight performance variation across contexts
- identify areas of systemic weakness

- provide insights into how AI systems behave outside high-resource environments

The results are therefore interpretive and comparative, rather than deterministic.

9 EXPECTED OUTCOMES

The Bharat AI Index is expected to reveal:

- performance gaps across regions and contexts
- differences in system behaviour across models
- inconsistencies in safety mechanisms
- limitations of current evaluation frameworks

10 SIGNIFICANCE

This framework contributes to:

- AI safety research
- evaluation of frontier models
- understanding of global deployment challenges

It also provides a starting point for:

- region-specific benchmarking
- policy discussions around equitable AI deployment